

- 1.** React is a JavaScript library used to build user interfaces. It helps developers create fast, interactive, and reusable components for web applications, especially Single Page Applications (SPAs).
- 2.** JSX (JavaScript XML) is a syntax extension for JavaScript that allows developers to write HTML-like code inside JavaScript. One advantage of JSX is that it makes the code more readable and easier to understand.
- 3.** Props are used to pass data from a parent component to a child component and cannot be modified by the child. State is used to store and manage data within a component, and it can be updated, causing the component to re-render.
- 4.** The `useState()` hook is used to create and manage state in a functional component. Example: `const [count, setCount] = useState(0);` Here, `count` stores the current value, and `setCount()` updates it.
- 5.** The `useEffect()` hook is used to perform side effects in React components. It is commonly used for fetching data from APIs, updating the document title, setting up timers, and adding or removing event listeners.
- 6.** The Virtual DOM is a lightweight copy of the real DOM. React compares the updated Virtual DOM with the previous version and updates only the changed parts of the real DOM, reducing unnecessary DOM manipulations and improving performance.
- 7.** The `key` prop uniquely identifies each item in a list. It helps React efficiently update, add, or remove list items.
- 8.** The `useRef()` hook is used to access DOM elements or store mutable values without causing a re-render. A common use case is automatically focusing an input field.
- 9.** The `useMemo()` hook memorizes a computed value, while `useCallback()` memorizes a function. `useMemo()` returns a value, whereas `useCallback()` returns a function.
- 10.** The Context API allows data to be shared across multiple components without prop drilling. It is commonly used for global data such as authentication, themes, language settings, and user preferences.